

A P/NP-Complete Classification of the Tait-Rolfsen Knot Table

K. Alimova, R. Karpenko, N. Lawrence, F. Rahimi, B. Sultangazieva, and T. Sultani
Adviser: Robert McGrail

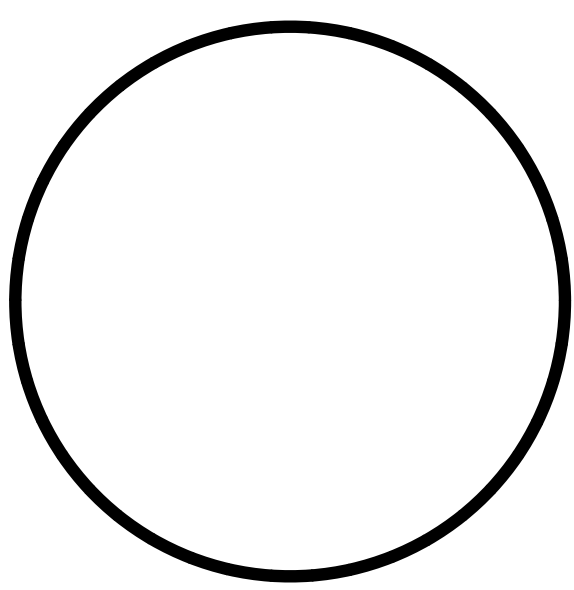
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Abstract

The goal of this project is to determine the nature of the CSPs over the Tait-Rolfsen knot table, the first 250 knots. At this time the only known polynomial time knot is the Unknot. Most of the rest of the knots of this table have been shown to be NP-Complete. This was achieved for each of these knots by coloring by an NP-Complete quandle. The quandle used are a set of so-called Sharac quandles.

Unknot

- The only known polynomial time knot is the Unknot.
- All CSPs over this knot are CSPs over 1 element sets.



Quandle U_2

- SAT is a CSP over U_2 .
- SAT is an NP-Complete CSP.
- SAT is a CSP over any Q containing U_2 .
- Any Q containing U_2 is NP-Complete.

* , /	0	1
0	0	0
1	1	1

Sharac Quandles

- Sharac j subquandle of j -cycles in S_j .
- Sharac j has subquandles of size 1 and 2 and nothing else.
- Sharac j is an NP-Complete algebra for $n \geq 4$.
- All colorings by Sharac j either trivial or full coloring.
- If a knot has a non-trivial coloring by Sharac j it is NP-Complete.

Sharac4

*	0	1	2	3	4	5
0	0	0	3	5	2	4
1	1	1	4	2	5	3
2	4	3	2	0	1	2
3	2	5	1	3	3	0
4	5	2	0	4	4	1
5	3	4	5	1	0	5

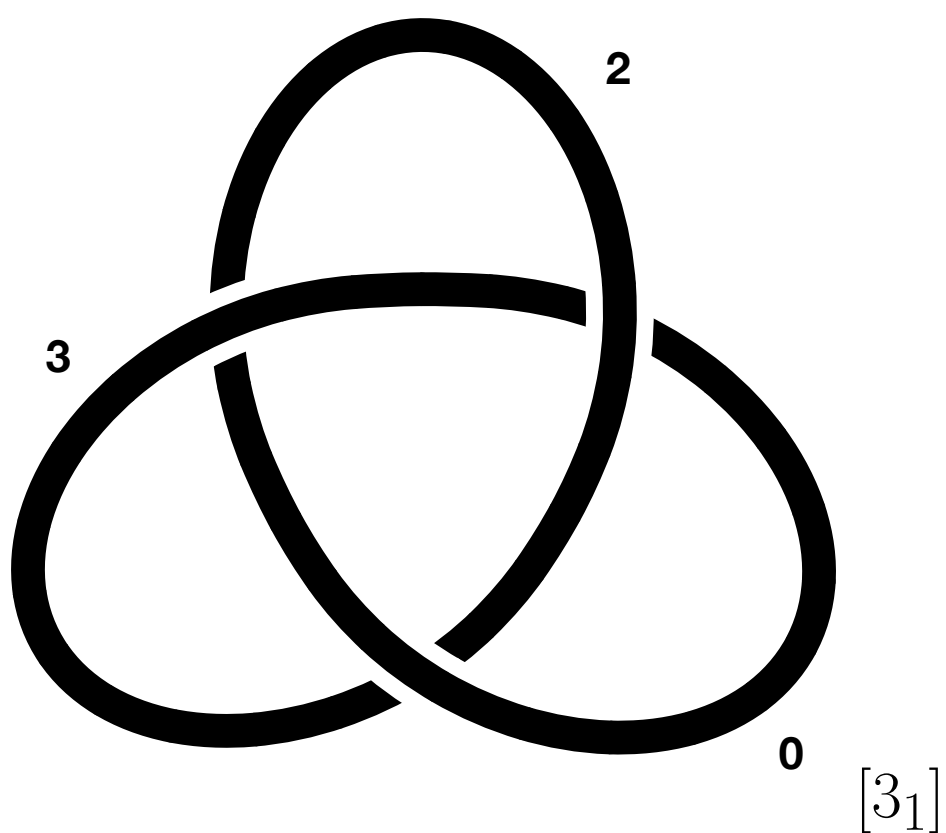
/	0	1	2	3	4	5
0	0	0	4	2	5	3
1	1	1	3	5	2	4
2	3	4	2	1	0	2
3	5	2	0	3	3	1
4	2	5	1	4	4	0
5	4	3	5	0	1	5

Known NP-Complete Knots

Quandle	Number of Knots
Sharac4	95
Sharac5	72
Sharac6	32
Sharac7	31
Sharac8	6
Sharac9	5

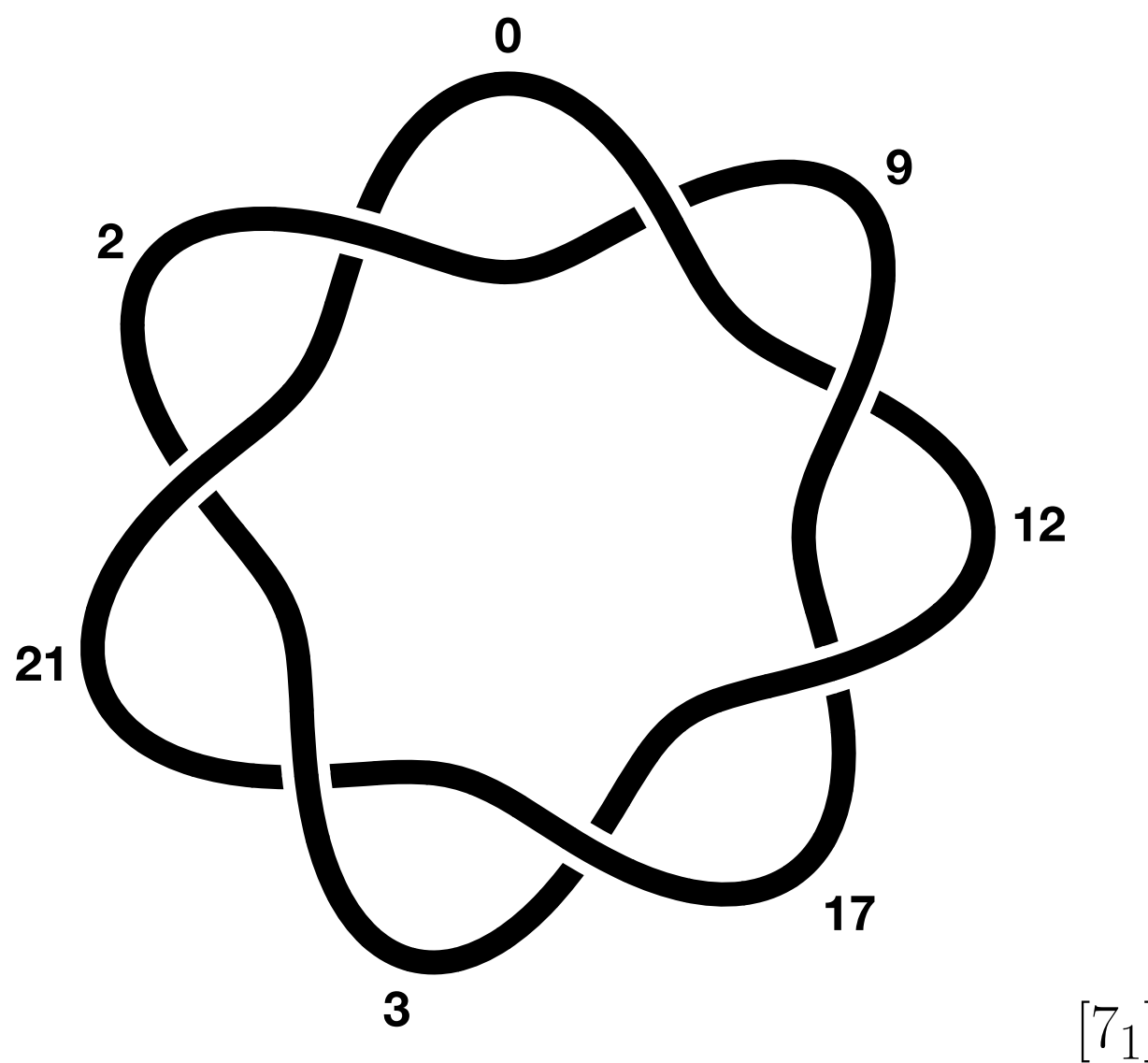
Sharac4 Colorable Knots

- $3_1, 6_1, 7_4, 7_7$
- $8_5, 8_{10}, 8_{11}, 8_{15}, 8_{18}, 8_{19}, 8_{20}, 8_{21}$
- $9_1, 9_2, 9_4, 9_6, 9_{10}, 9_{11}, 9_{15}, 9_{16}, 9_{17}, 9_{23}, 9_{24}, 9_{28}, 9_{29}, 9_{34}, 9_{35}, 9_{37}, 9_{38}, 9_{40}, 9_{46}, 9_{47}, 9_{48}$
- $10_4, 10_5, 10_9, 10_{10}, 10_{14}, 10_{19}, 10_{21}, 10_{29}, 10_{31}, 10_{32}, 10_{36}, 10_{40}, 10_{42}, 10_{59}, 10_{61}, 10_{62}, 10_{63}, 10_{64}, 10_{65}, 10_{66}, 10_{67}, 10_{68}, 10_{69}, 10_{74}, 10_{75}, 10_{76}, 10_{77}, 10_{78}, 10_{82}, 10_{84}, 10_{85}, 10_{87}, 10_{89}, 10_{96}, 10_{97}, 10_{98}, 10_{99}, 10_{103}, 10_{106}, 10_{107}, 10_{108}, 10_{112}, 10_{113}, 10_{114}, 10_{120}, 10_{122}, 10_{136}, 10_{139}, 10_{140}, 10_{141}, 10_{142}, 10_{143}, 10_{144}, 10_{145}, 10_{146}, 10_{147}, 10_{158}, 10_{159}, 10_{160}, 10_{163}, 10_{164}, 10_{165}$



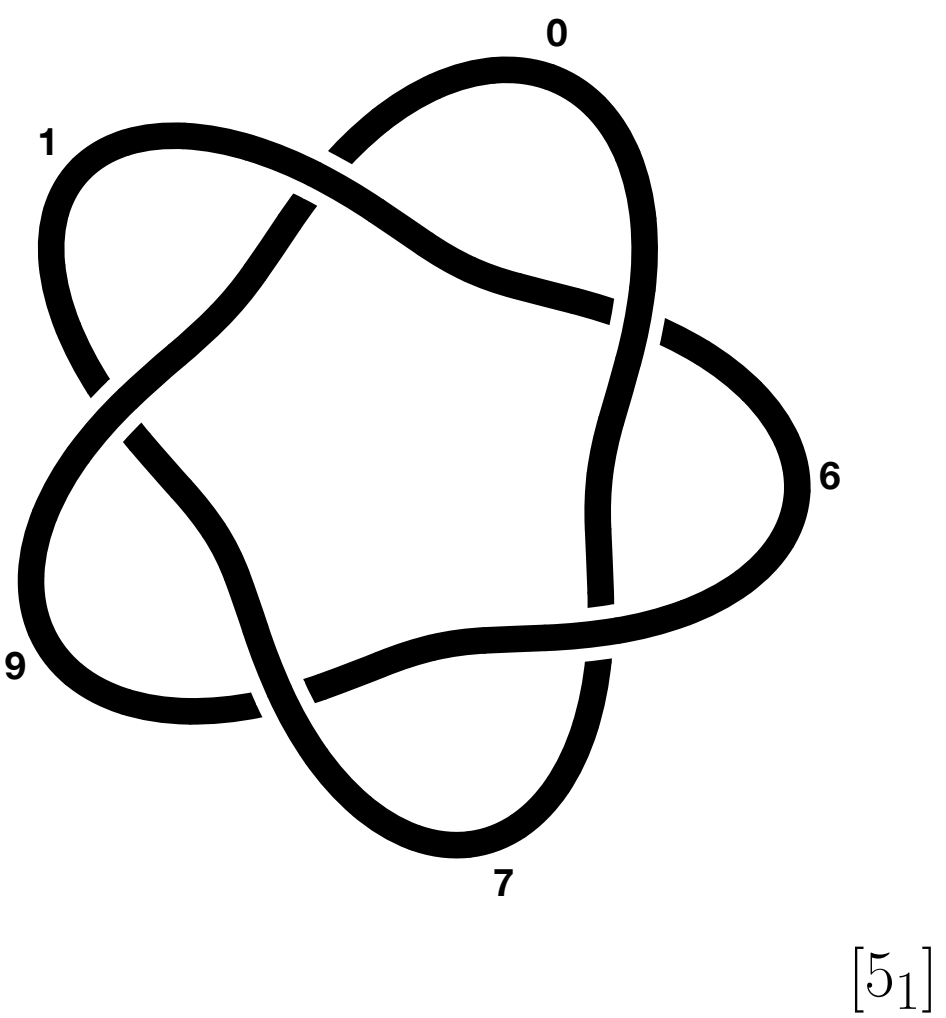
Sharac7 Colorable Knots

- $7_1, 7_2, 7_6$
- 8_{13}
- $9_7, 9_9, 9_{22}, 9_{25}, 9_{30}, 9_{36}, 9_{42}, 9_{45}$
- $10_3, 10_{12}, 10_{22}, 10_{23}, 10_{24}, 10_{39}, 10_{71}, 10_{72}, 10_{83}, 10_{86}, 10_{90}, 10_{92}, 10_{93}, 10_{95}, 10_{102}, 10_{104}, 10_{115}, 10_{118}, 10_{119}$



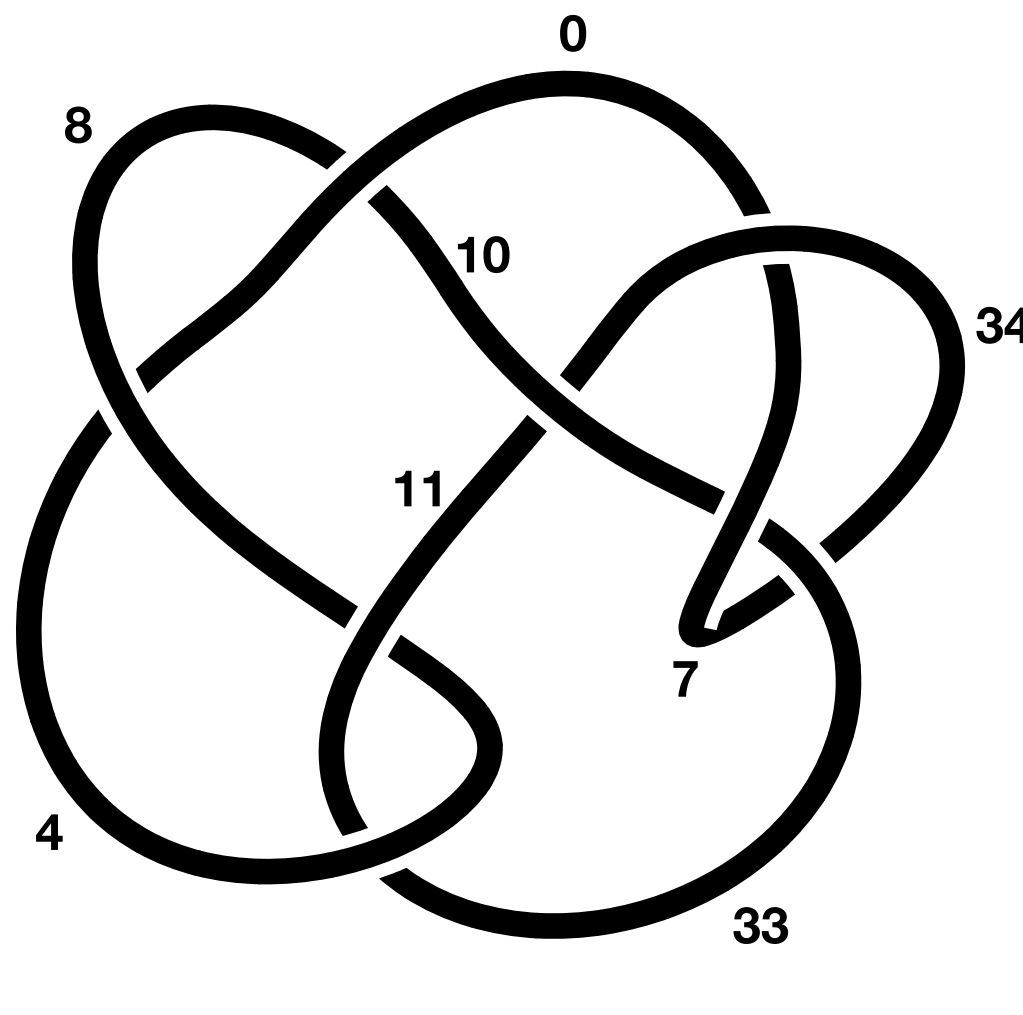
Sharac5 Colorable Knots

- $5_1, 5_2$
- $8_1, 8_3, 8_4, 8_6, 8_7, 8_8$
- $9_5, 9_{13}, 9_{14}, 9_{20}, 9_{26}, 9_{31}, 9_{39}, 9_{41}, 9_{49}$
- $10_1, 10_2, 10_7, 10_{13}, 10_{16}, 10_{20}, 10_{25}, 10_{26}, 10_{27}, 10_{30}, 10_{34}, 10_{43}, 10_{46}, 10_{47}, 10_{48}, 10_{49}, 10_{50}, 10_{51}, 10_{52}, 10_{53}, 10_{54}, 10_{55}, 10_{56}, 10_{57}, 10_{79}, 10_{80}, 10_{81}, 10_{100}, 10_{101}, 10_{111}, 10_{116}, 10_{123}, 10_{124}, 10_{125}, 10_{126}, 10_{127}, 10_{128}, 10_{129}, 10_{130}, 10_{131}, 10_{132}, 10_{133}, 10_{134}, 10_{135}, 10_{148}, 10_{149}, 10_{150}, 10_{151}, 10_{152}, 10_{153}, 10_{154}, 10_{155}, 10_{156}, 10_{157}, 10_{162}$



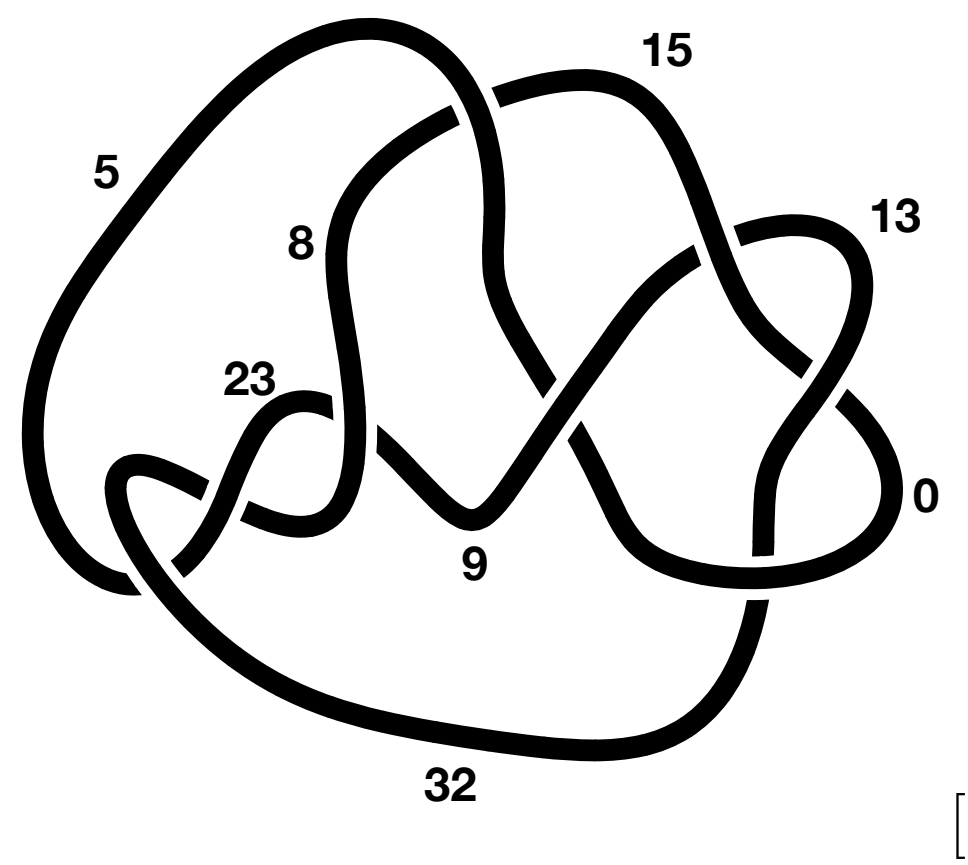
Sharac8 Colorable Knots

- $8_{14}, 9_{18}, 9_{27}$
- $10_{15}, 10_{33}, 10_{37}$



Sharac9 Colorable Knots

- $8_9, 9_{19}$
- $10_{17}, 10_{41}, 10_{45}$



List of Uncolored Knots

- $6_3, 7_5, 8_{12}, 8_{17}$
- $9_{21}, 9_{32}$
- $10_{44}, 10_{88}$